LAB EXPERIMENT RECORD

SET 21: ENERGY CONSUMPTION ANALYSIS

Department	AIDS
Course Code / Course Name	DSA0601 / Data Handling and Visualization
Experiment	Data Visualization using R
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Aim

To analyze energy consumption data using a histogram, density plot, bubble scatter plot, and bar chart, and to build an interactive dashboard using R and Shiny.

Dataset: Monthly Sales

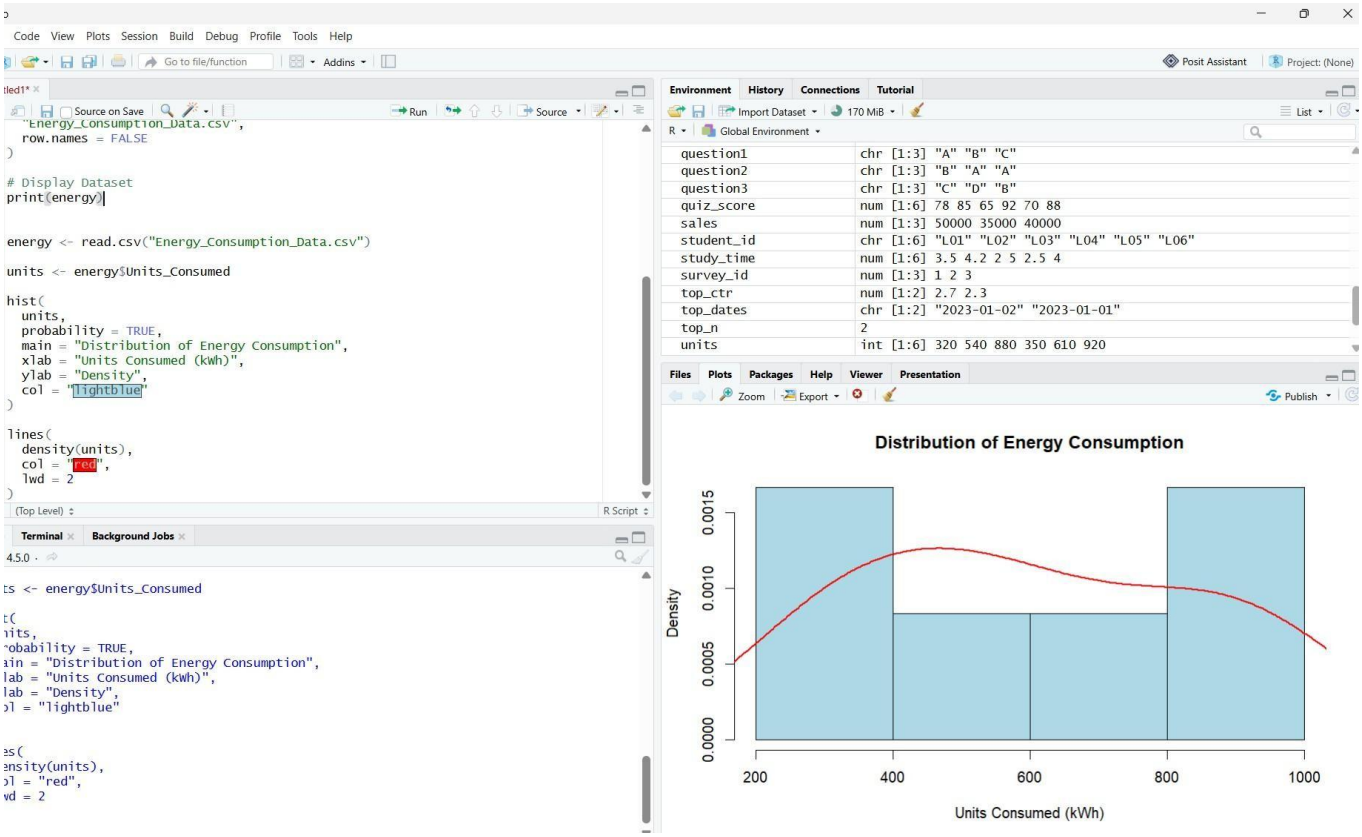
Sector	Region	Month	Temperature (°C)	Units Consumed (kWh)	Cost (₹)	Renewable Usage (%)
Residential	North	Jan	15	320	2100	22
Commercial	South	Jan	24	540	3600	18
Industrial	West	Feb	20	880	5900	12
Residential	East	Feb	18	350	2300	25
Commercial	North	Mar	28	610	4100	20

PROGRAM 1: Histogram and Density Plot of Energy Consumption

To visualize the distribution of units consumed using a histogram and density plot.

```
# # Import Dataset
energy <- read.csv("Energy_Consumption_Data.csv")
units <- energy$Units_Consumed
# Histogram
hist(
  units,
  probability = TRUE,
  main = "Distribution of Energy Consumption",
  xlab = "Units Consumed (kWh)",
  ylab = "Density",
  col = "lightblue"
)
```

```
# Density Plot
lines(
  density(units),
  col = "red",
  lwd = 2
)
```

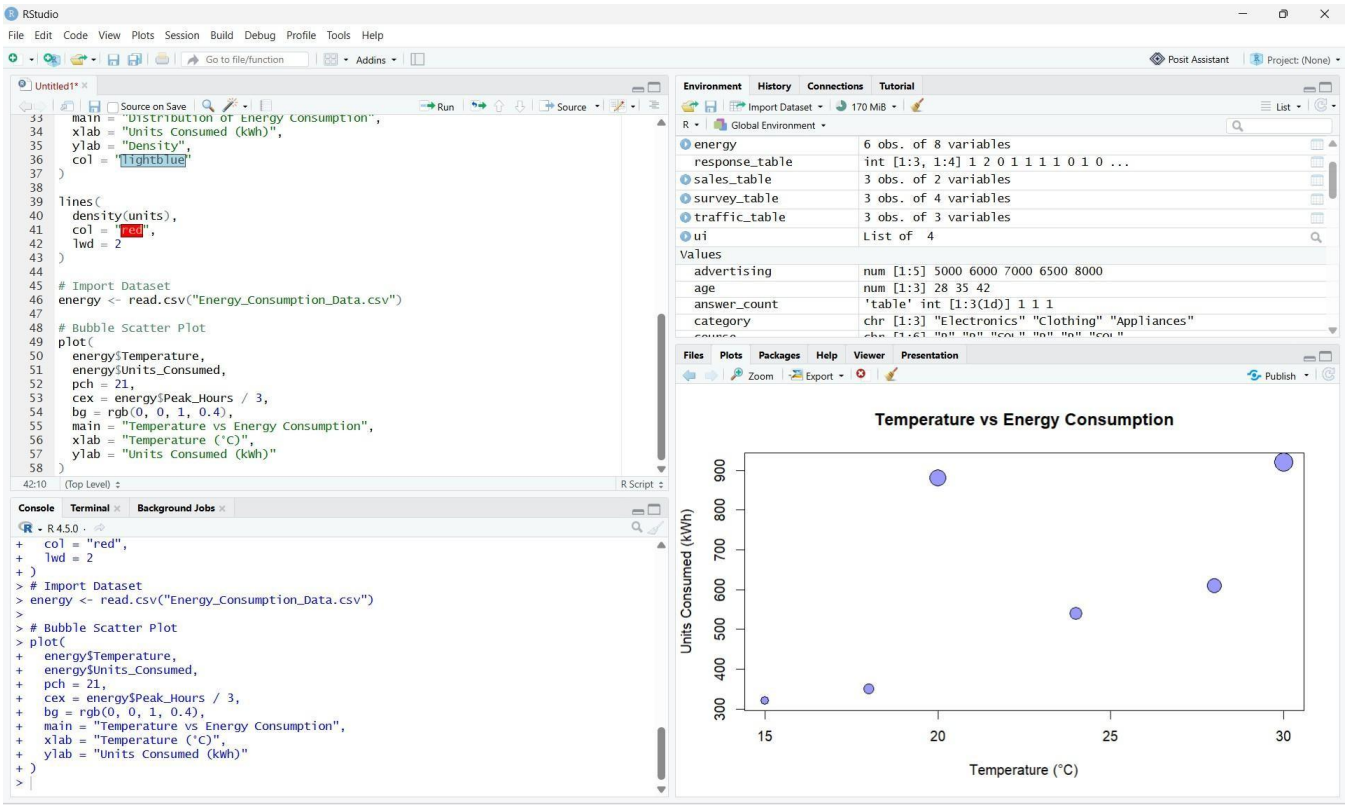


Observation: Energy consumption ranges from 320 to 920 kWh. The histogram displays the frequency distribution, while the red density curve shows the overall pattern of consumption.

PROGRAM 2: Bubble Scatter Plot of Temperature vs Energy Consumption

To analyze the relationship between temperature and units consumed, using bubble size to represent peak hours and transparency to improve visualization

```
# Import Dataset
energy <- read.csv("Energy_Consumption_Data.csv")
# Bubble Scatter Plot
plot(
  energy$Temperature,
  energy$Units_Consumed,
  pch = 21,
  cex = energy$Peak_Hours / 3,
  bg = rgb(0, 0, 1, 0.4),
  main = "Temperature vs Energy Consumption",
  xlab = "Temperature (°C)",
  ylab = "Units Consumed (kWh)"
)
```

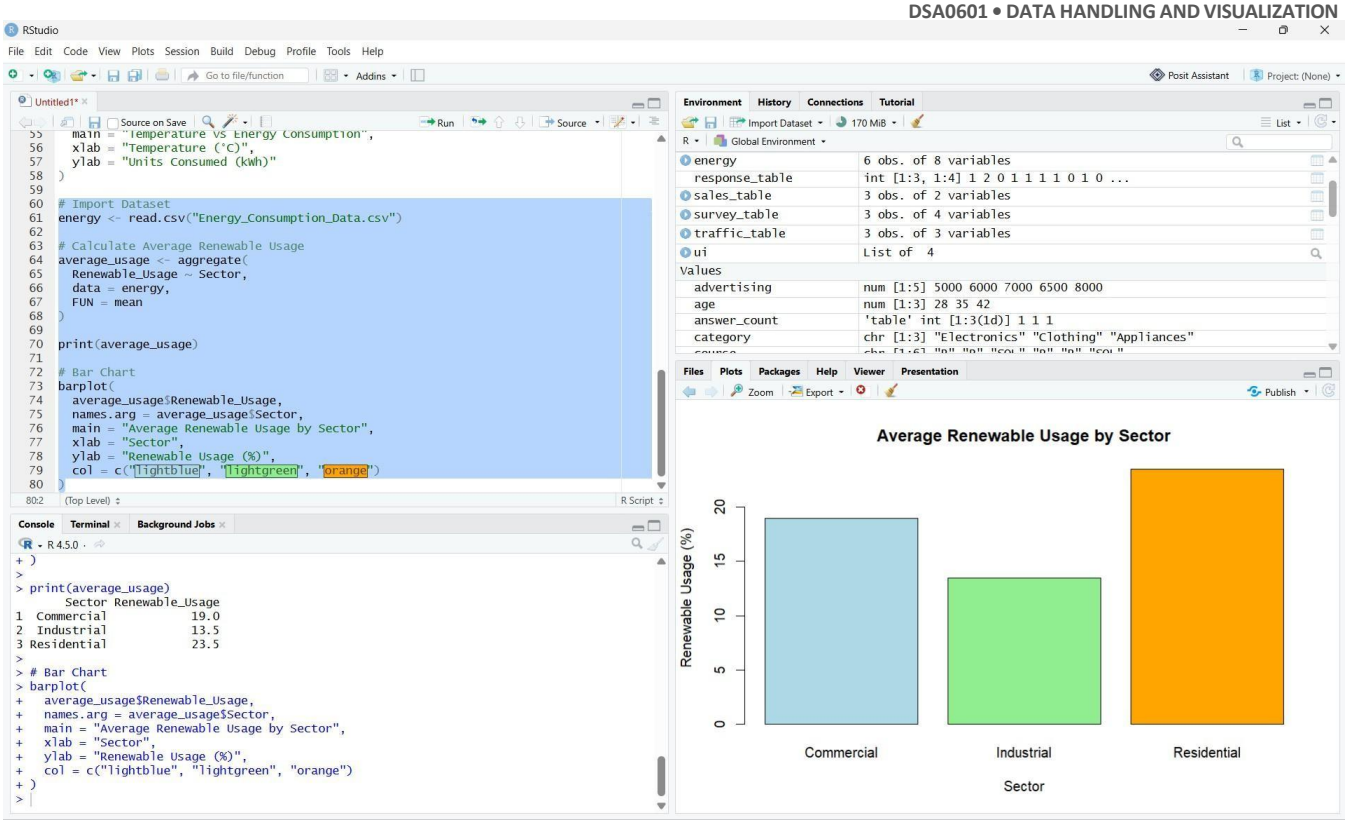


Observation: Industrial observations have the highest energy consumption and larger bubbles because they have more peak hours. The six records alone are insufficient to establish a strong general relationship between temperature and energy consumption.

PROGRAM 3: Average Renewable Usage by Sector

To calculate the average renewable energy usage for each sector and visualize it using a bar chart.

```
# Import Dataset
energy <- read.csv("Energy_Consumption_Data.csv")
# Calculate Average Renewable Usage
average_usage <- aggregate(
  Renewable_Usage ~ Sector,
  data = energy,
  FUN = mean
)
print(average_usage)
# Bar Chart
barplot(
  average_usage$Renewable_Usage,
  names.arg = average_usage$Sector,
  main = "Average Renewable Usage by Sector",
  xlab = "Sector",
  ylab = "Renewable Usage (%)",
  col = c("lightblue", "lightgreen", "orange")
)
```



Insight: Residential has the highest average renewable usage at 23.5%, while Industrial has the lowest at 13.5%. This indicates greater renewable energy adoption in the residential records provided.

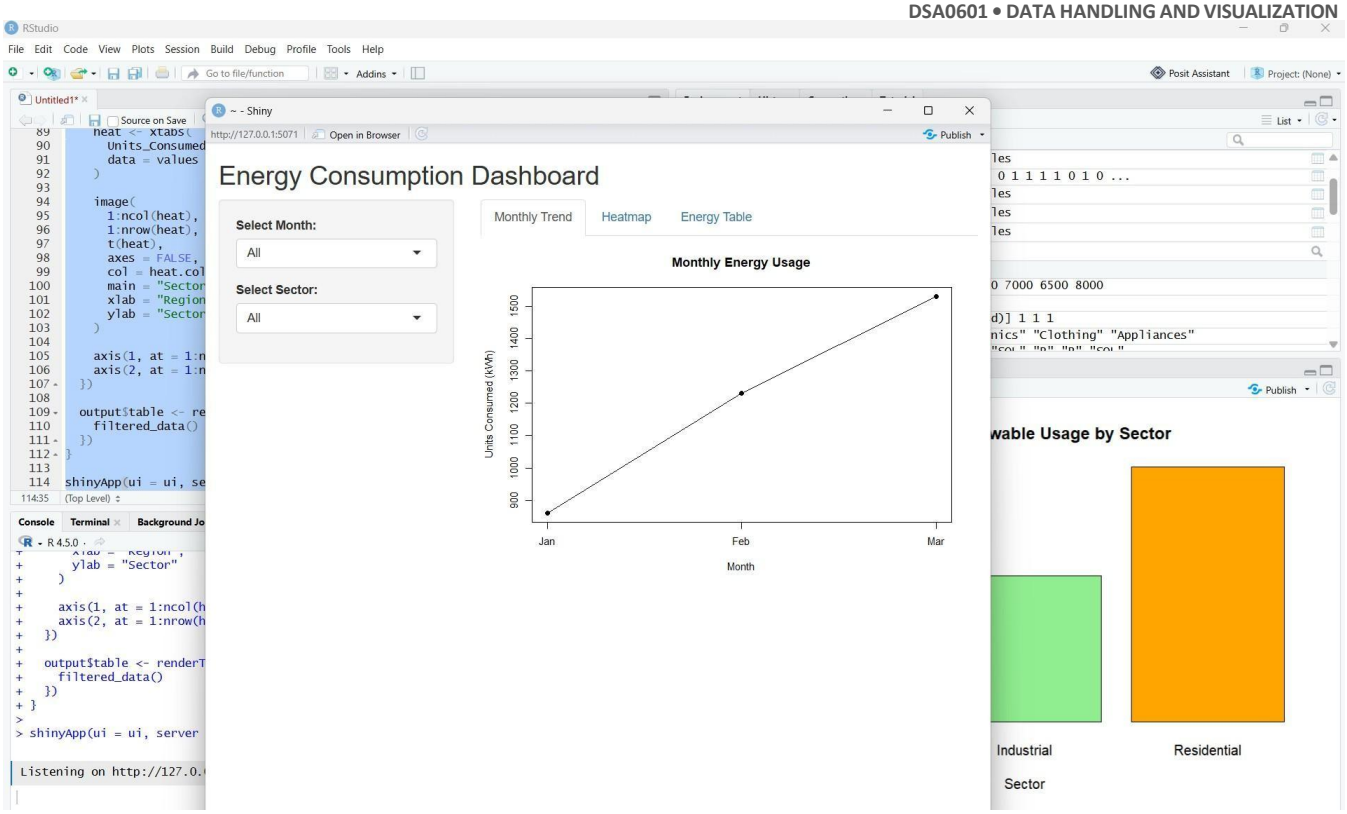
PROGRAM 4: Interactive Energy Consumption Dashboard

To build an interactive dashboard combining a monthly energy usage line chart, a sector-region heatmap, and a data table using R and Shiny

```
library(shiny)
energy <- read.csv("Energy_Consumption_Data.csv")
ui <- fluidPage(
  titlePanel("Energy Consumption Dashboard"),
    mainPanel(
      tabsetPanel(
        tabPanel("Monthly Trend", plotOutput("trend")),
        tabPanel("Heatmap", plotOutput("heatmap")),
        tabPanel("Energy Table", tableOutput("table"))
      )
    )
  )

server <- function(input, output) {
  filtered_data <- reactive({
    d <- energY

    axis(2, at = 1:nrow(heat), labels = rownames(heat))
  })
  output$table <- renderTable({
    filtered_data()
  })
}
shinyApp(ui = ui, server = server)
```



Observation: The dashboard allows users to filter energy data by month and sector, view monthly consumption trends, and compare consumption across sectors and regions.

Procedure

1. Import the Energy_Consumption_Data.csv dataset into RStudio.
2. Extract the Units_Consumed column and create a histogram with a density curve.
3. Use plot() to create a bubble scatter plot between temperature and units consumed, representing peak hours using bubble size.
4. Calculate average renewable usage for each sector using aggregate() and display the results in a bar chart.
5. Load the Shiny package and develop an interactive dashboard with month and sector filters, a line chart, heatmap, and table.
6. Run the dashboard using Run App and explore the visualizations.

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Result

The energy consumption dataset was successfully analyzed using a histogram, density plot, bubble scatter plot, and bar chart. An interactive Shiny dashboard was also developed to explore monthly energy consumption and renewable energy usage.